Line Balancing Study - Assembly Line 2 - Section 1: Executive Summary & Scope

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck. Secondary control 1.2: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.

- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored

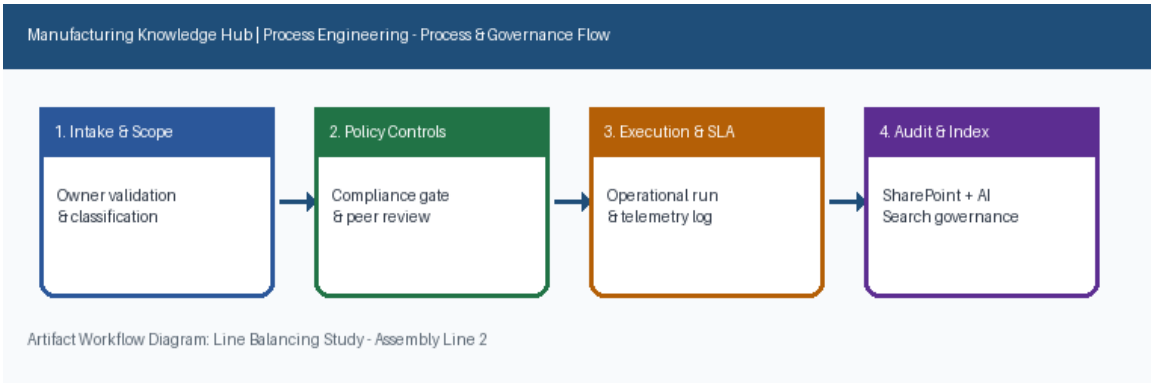
Line Balancing Study - Assembly Line 2 - Section 2: Architecture & Process Flow

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt. Secondary control 2.2: Projected throughput improvement is approximately 6% without added headcount.

- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Projected throughput improvement is approximately 6% without added headcount.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-2	Carlos Ibarra	3 business days	Stage 2 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 3: Governance Standards & Controls

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Projected throughput improvement is approximately 6% without added headcount. Secondary control 3.2: Recommendation is scheduled for pilot validation next production cycle.

- Projected throughput improvement is approximately 6% without added headcount.
- Recommendation is scheduled for pilot validation next production cycle.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-3	Carlos Ibarra	10 business days	Stage 3 Verified

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

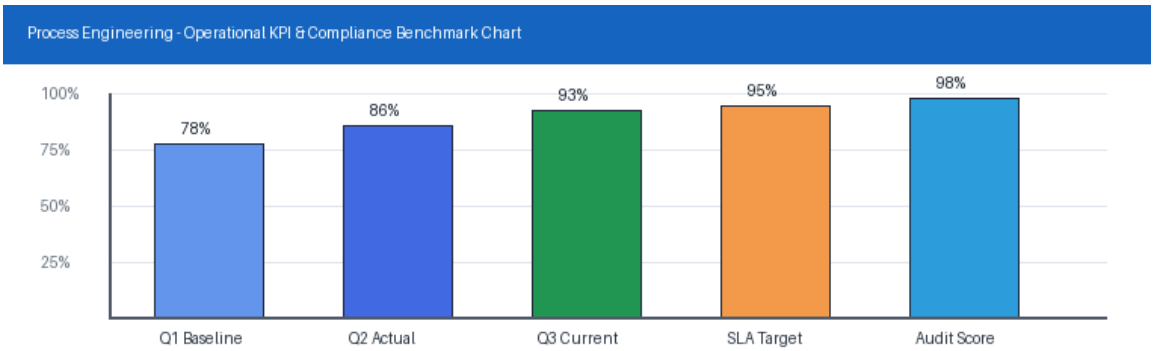
Line Balancing Study - Assembly Line 2 - Section 4: Operational KPIs & Benchmarks

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 4 (Operational KPIs & Benchmarks) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Tracks baseline performance, target SLAs, and quarterly benchmark progression.

Operational requirement 4.1: Recommendation is scheduled for pilot validation next production cycle. Secondary control 4.2: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.

- Recommendation is scheduled for pilot validation next production cycle.
- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-4	Carlos Ibarra	5 business days	Stage 4 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 5: Roles, Responsibilities & RACI

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 5 (Roles, Responsibilities & RACI) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Clarifies ownership across accountable executives, operational leads, and reviewers.

Operational requirement 5.1: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck. Secondary control 5.2: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.

- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Section 5 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-5	Carlos Ibarra	3 business days	Stage 5 Verified

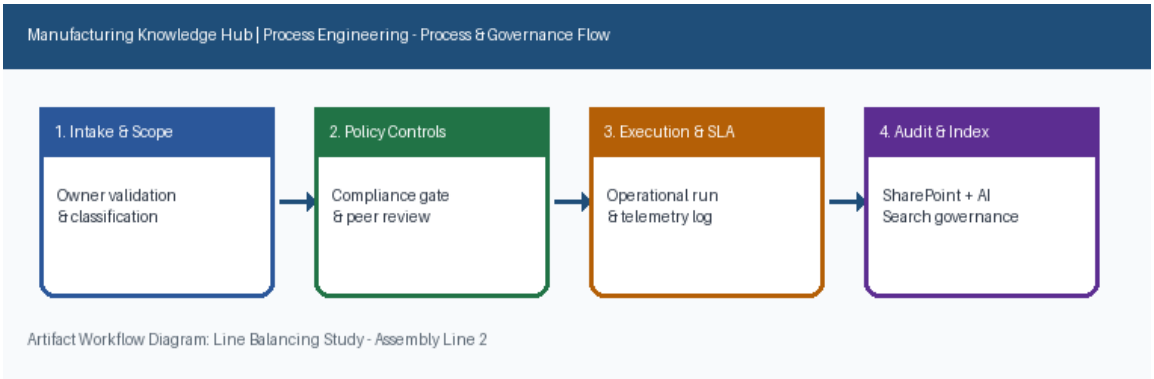
Line Balancing Study - Assembly Line 2 - Section 6: Standard Operating Procedures

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 6 (Standard Operating Procedures) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Provides step-by-step execution checklists, intake criteria, and verification gates.

Operational requirement 6.1: Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt. Secondary control 6.2: Projected throughput improvement is approximately 6% without added headcount.

- Redistributing one sub-assembly step to Station 3 reduced the bottleneck to within 2 seconds of takt.
- Projected throughput improvement is approximately 6% without added headcount.
- Section 6 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-6	Carlos Ibarra	10 business days	Stage 6 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Line Balancing Study - Assembly Line 2 - Section 7: Risk Register & Mitigations

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 7 (Risk Register & Mitigations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Catalogs operational, compliance, and delivery risks alongside preventive mitigations.

Operational requirement 7.1: Projected throughput improvement is approximately 6% without added headcount. Secondary control 7.2: Recommendation is scheduled for pilot validation next production cycle.

- Projected throughput improvement is approximately 6% without added headcount.
- Recommendation is scheduled for pilot validation next production cycle.
- Section 7 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-7	Carlos Ibarra	5 business days	Stage 7 Verified

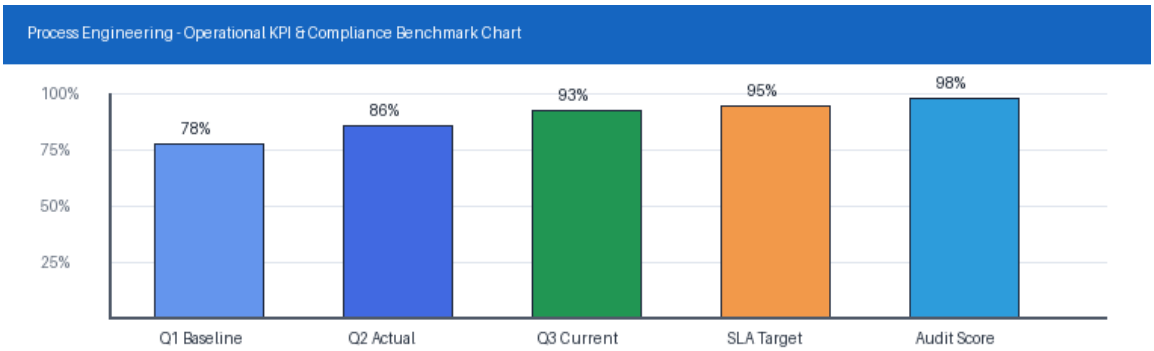
Line Balancing Study - Assembly Line 2 - Section 8: Appendix, Lineage & Sign-off

Time study and line balancing analysis identifying the bottleneck station and proposed cycle time redistribution. Section 8 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 8.1: Recommendation is scheduled for pilot validation next production cycle. Secondary control 8.2: Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.

- Recommendation is scheduled for pilot validation next production cycle.
- Station 4 cycle time exceeded takt time by 8 seconds, creating a recurring bottleneck.
- Section 8 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Line Balancing Study - Assem Gate 1	Carlos Ibarra	3 business days	Active
Line Balancing Study - Assem Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-8	Carlos Ibarra	3 business days	Stage 8 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online